



SRM
INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be **University** *u/s 3 of UGC Act, 1956)*

AI-Driven Discovery Hub

Powered by

Sentient Scripts Private Limited

In association with

Department of Computational Intelligence
School of Computing

Internship Report

Enzo JUHEL

SENTIENT SCRIPTS

SRM INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

AI-DRIVEN DISCOVERY HUB

Industry Sponsored Lab
Powered by **Sentient Scripts**
in collaboration with **the Department of CINTEL**, SRMIST KTR

12/08/2024

TO WHOMEVER IT MAY CONCERN

This is to certify that **Mr ENZO JUHEL from Epita University, France** has completed his internship at **AI-Driven Discovery Hub, Department of Computational Intelligence, SRM Institute of Science and Technology** from **18/01/2024** to **12/08/2024**. He has actively participated in the internship throughout his tenure.

During his tenure with us for the above period, we found him efficient, working in technologies like Computer vision and Generative AI.

We wish **ENZO JUHEL** the very best in his future endeavours.

Best Regards

Dr.R.Udendhran
Assistant Professor
Department of Computational Intelligence

Dr.R.Annie Uthra
Professor & Head,
Department of Computational Intelligence

COLLABORATORS AND SUPERVISORS

SRMIST Core Members:

- Dr. R. Annie Uthra
- Dr. Anupama CG
- Dr. R. Udendhran
- Dr. G. Dinesh
- Dr. Selvakumarasamy S
- Dr. R. Babu

Sentient Scripts Members:

- Mr. Naveen Murali
- Ms. Priya George

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EPITA ÉCOLE D'INGÉNIEURS EN INFORMATIQUE
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1 Introduction

As part of my international semester at SRM University in Chennai, India, I had the opportunity to undertake a five-month internship as a full-stack developer with the AI-Driven Discovery Hub. This internship allowed me to immerse myself in a professional environment while continuing my academic studies. The AI-Driven Discovery Hub, under the guidance of Dr. R Udendhran, connected me with Sentient Scripts, a company specializing in innovative software solutions, where I was assigned to work on two significant projects.

During this internship, I was responsible for developing full-stack applications using Next.js for the front-end and a Python-based API for the back-end. The experience provided me with hands-on exposure to the entire development process, from initial concept and design to final implementation and deployment. It also allowed me to deepen my understanding of modern web technologies and AI tools, which are crucial in today's tech-driven landscape.

This report will detail my experiences, the challenges I faced, and the solutions I implemented throughout the internship. Additionally, it will highlight the skills I acquired and how this internship has contributed to my professional and personal growth. Through this opportunity, I was able to bridge the gap between theoretical knowledge and practical application, making this internship an essential part of my education and future career.

2 Internship progress

2.1 April

My first month at the AI-Driven Discovery Hub has been both challenging and rewarding. I was assigned an initial project that required the use of Python to manipulate various AI tools and frameworks. This project introduced me to several key concepts and tools, including Rasa, NLU, Gamma, Mistral, and Langchain, all of which I needed to master in order to effectively handle AI in the context of the project.

2.1.1 Learning the Basics

The first step in my journey was to familiarize myself with the vocabulary and fundamental concepts of artificial intelligence. This was crucial as it laid the groundwork for understanding how different AI components interact and function. I spent a considerable amount of time studying and comprehending terms and principles, which was essential for my progress.

2.1.2 Exploring Rasa

Once I had a solid grasp of the AI basics, I delved into learning about Rasa. Rasa is a popular open-source framework for building conversational AI, and it played a central role in my project. My tasks included optimizing, customizing, and fine-tuning the AI to meet the project's specific goals. This involved extensive research and hands-on experimentation to understand how Rasa's NLU (Natural Language Understanding) and other components work.

2.1.3 Discovering Ollama and Mistral

After weeks of research and trials, I discovered Ollama, which proved to be more suitable for the project's needs. Ollama's capabilities allowed me to utilize Mistral more effectively for multiple applications. The Python API provided by Ollama enabled me to write my first useful lines of code for the project. This API facilitated the integration of Mistral, allowing for a more seamless and efficient development process.

2.1.4 Project Implementation

Using the tools and the knowledge I had acquired, I began implementing the project. This involved not only coding but also continuous learning and adaptation. Each step presented new challenges and learning opportunities, from debugging code to refining the AI's responses. The combination of Rasa's robust framework and Ollama's user-friendly API significantly streamlined the development process, enabling me to achieve tangible results.

In summary, my first month at the AI-Driven Discovery Hub has been an intense learning experience. I have gained valuable insights into AI technologies and frameworks, particularly Rasa and Ollama. The knowledge and skills I have acquired during this period have laid a strong foundation for my future work in AI and machine learning. I look forward to continuing my journey, tackling new challenges, and contributing to innovative AI-driven projects.

2.2 May

During my second month of internship in May, I continued the development of the initial project, dividing it into two main components: the front-end and the back-end.

2.2.1 Back-End Development

For the back-end, I focused on learning how to retrieve information from the internet using Python libraries. Then how to create customized models for various applications with Ollama and developing a Flask API in Python to facilitate communication with the web interface. This experience significantly enhanced my understanding of backend processes and API integration, which are crucial for developing robust and efficient web applications.

2.2.2 Front-End Development

Being my first foray into front-end development, I initially created a web interface using HTML and CSS to interact with my Flask API. This foundational work helped me grasp the basics of web development. However, for the final project, I needed to learn the fundamentals of Next.js and TypeScript. To do this, I started by creating a simple "to-do list" application. This exercise was instrumental in helping me acquire the necessary skills for a Next.js project that utilizes a Flask API with Python.

I then modified the "to-do list" application to send the text of a new task to my API and add the API response to the list of tasks. This modification introduced me to more complex interactions between the front-end and back-end. However, I encountered a significant issue where the delay from my API caused an error code in my Next.js interface. This challenge required me to delve deeper into debugging and optimizing the interaction between the front-end and back-end components.

2.2.3 Exploring Vercel

To overcome some of the hurdles I faced, I discovered Vercel, a platform that provides open-source interfaces for inspiration. I learned to use Vercel to copy, test, and deploy Next.js projects. This exploration allowed me to see how professional-level projects are structured and implemented, giving me valuable insights and practical knowledge that I could apply to my own work.

May was a month of substantial growth and learning. I deepened my expertise in back-end development, particularly with Python APIs and Flask, while also venturing into the world of front-end development with HTML, CSS, Next.js, and TypeScript. Despite encountering challenges, such as API delays causing errors in the front-end, I learned valuable lessons in problem-solving and optimizing web applications. My experience with Vercel further broadened my understanding of modern web development practices. I look forward to continuing this journey and applying what I have learned to complete the project successfully.

2.3 June

In June, I began working on the second project assigned by Sentient Scripts while continuing to make progress on the first project.

2.3.1 Exploring and Implementing YOLOv8

For the second project, I initially focused on studying, mastering, and coding with the YOLOv8 tool to detect specific predefined elements in images using Python. However, I soon encountered the limitations of YOLOv8 in accurately detecting the elements required for this project. This led me to search for alternative tools that could provide better accuracy.

2.3.2 Discovering mmfashion

During my search, I discovered mmfashion, an open-source project designed to describe a specific category of elements accurately using PyTorch. To effectively use mmfashion, I had to locate appropriate models and datasets online to achieve reliable results. I then analyzed the code to understand the logic used and adapted it for the second project. This process involved significant research and hands-on experimentation to ensure mmfashion could meet the project's needs.

2.3.3 Enhancing the First Project

In parallel, I continued to work on the first project with Ollama. My goal was to find a way to reduce the latency when sending data from the API without compromising the model's accuracy. I discovered the technique of sending data in chunks, where each word generated by Mistral is sent immediately. This method significantly improved the response time.

I first implemented this chunking method for an HTML interface and, after achieving promising results, tested it on a Next.js interface. Through my research on Vercel, I found an open-source interface that perfectly suited the project's needs. I integrated the necessary components from this interface into my own and incorporated my Flask API to stream Mistral's responses.

2.3.4 Finalizing the First Project

By the end of June, I successfully finalized the first project. The interface now efficiently handles API responses with reduced latency, thanks to the chunking method. This achievement was a significant milestone, demonstrating my ability to innovate and improve project performance through research and adaptation.

June was a month of intensive learning and development. I expanded my skills in using AI tools like YOLOv8 and mmfashion and enhanced my understanding of latency reduction techniques. Balancing two projects simultaneously, I managed to complete the first project and made substantial progress on the second. These experiences have greatly enriched my technical knowledge and problem-solving abilities, preparing me for more complex challenges in the future.

2.4 July

At the beginning of July, I created a private GitHub repository to host the second project, leveraging the advantages of version control and branching. This allowed for better organization and management of the project's development.

2.4.1 Mastering Docker

To ensure the Python API ran in a controlled and replicable environment, I dedicated time to mastering Docker. I studied various YouTube tutorials to set up a workspace that combined Visual Studio Code and Docker, complete with an integrated debugger. This setup provided a robust environment for developing and testing the API. Please refer to the code in the appendix at the end of this report for additional information.

2.4.2 Overcoming NixOS Challenges

As I use NixOS on my computer, I experimented with using a 'shell.nix' file to manage the necessary Python dependencies. Due to the declarative nature of NixOS, python virtual environments must not be used. I built certain versions of Python dependencies and utilized those available in the NixOS package manager. Unfortunately, the built dependencies did not work, likely due to unspecified sub-dependencies. Consequently, I relied on Docker throughout the second project to run the Flask-based Python API. For further details, please refer to the code provided in the appendix at the end of this report.

2.4.3 Enhancing Next.js Skills

I further honed my Next.js skills by following online courses on YouTube, comprising four detailed videos, for a total of 10 hours. These tutorials equipped me with the knowledge to create a Next.js frontend application from scratch. Using the Shadcn framework, I accessed numerous modern and minimalist UI components. After defining the architecture of my interface, I successfully incorporated these components.

2.4.4 Overcoming Technical Challenges

While developing the frontend, I encountered some difficulties, particularly with an element that allowed for the uploading of local files to the application. Despite these challenges, I persevered and eventually resolved all issues, ensuring the smooth functioning of the application.

July was a month of significant learning and development. I gained proficiency in Docker, enabling the creation of a reliable environment for the Python API. I overcame the challenges posed by NixOS, ensuring seamless development despite the limitations. My enhanced understanding of Next.js and the successful integration of Shadcn components marked a major milestone in my front-end development skills. This month's work has solidified my technical capabilities and prepared me for future challenges in software development.

2.5 August

In August, I successfully completed the second project assigned to me by Sentient Scripts. After finalizing the project, I was granted access to a Microsoft Azure container provided by the company, where I securely uploaded both completed projects for further review and analysis by their team.

2.5.1 End of Internship

This final month marked the conclusion of my five-month internship, during which I applied and honed my skills in full-stack development, particularly in the areas of AI and software engineering. The completion of these projects not only allowed me to demonstrate my technical abilities but also to contribute to the company's ongoing research and development efforts.

2.5.2 Internship Certificate

Upon the successful completion of my internship, I received a certificate from Sentient Scripts and the AI-Driven Discovery Hub, recognizing my contributions and the skills I developed during my time with them. This internship has been an invaluable experience, providing me with practical knowledge and insights that will undoubtedly benefit my future career in the tech industry.

3 Environment

During my internship, I had the opportunity to work with Sentient Scripts, a company I was connected to through the AI-Driven Discovery Hub at SRM University, where I had been accepted after requesting to undertake an internship alongside my studies. Sentient Scripts is a company focused on cutting-edge AI technologies and software development, with a particular emphasis on innovative solutions that leverage artificial intelligence for various applications.

My involvement with the company was made possible through the AI-Driven Discovery Hub, a research center at SRM University, led by Dr. R. Udendhran. The hub serves as a bridge between academic research and industry applications, providing students and researchers with the opportunity to work on real-world projects.

During my time with Sentient Scripts, I worked on two significant projects that required a deep understanding of AI and software development, specifically in full-stack development. However, due to a confidentiality agreement, I am unable to share specific details or excerpts of the code or any other sensitive information related to the projects.

Sentient Scripts provided an enriching environment where I could apply and expand my technical skills, particularly in the development of AI-driven solutions, making this internship a critical part of my professional growth.

3.1 Professionals met

During my internship, I had the opportunity to meet and interact with several professionals who significantly contributed to my learning experience and the success of my projects.

3.1.1 Dr. R Udendhran

My journey began at the AI-Driven Discovery Hub of SRM University, where I met Dr. R Udendhran. He is an assistant professor and data scientist specializing in Computational Intelligence. Dr. Udendhran, provided me with the essential information for the first project. His guidance laid the foundation upon which I expanded my knowledge through further research on the internet. His expertise and willingness to share knowledge were invaluable to my progress.

3.1.2 Mr. Naveen Murali

I also had the privilege of interacting with Mr. Naveen Murali, a data scientist at Sentient Scripts, the company for which I completed two projects. Our communication was primarily through video conferences and email exchanges. Each week, I sent him a document summarizing my progress and outlining my plans for the upcoming week. His feedback and insights were crucial in steering my projects in the right direction.

3.1.3 Ms. Priya George

At the beginning of my internship, I also exchanged ideas with Ms. Priya George. She provided initial direction for both projects and answered numerous questions I had. Her early guidance helped me set a clear path for my work and ensured I was aligned with the project goals.

3.2 Collaboration with Fellow Students

In addition to the professionals, I also interacted with fellow students from SRM University who were part of the same internship program. These exchanges were highly enriching as we shared knowledge, resources, and experiences. Collaborating with peers added a collaborative dimension to my learning and helped me gain different perspectives on problem-solving.

Meeting and working with these professionals and students greatly enriched my internship experience. Their guidance, feedback, and support were instrumental in the completion of my projects and in the overall development of my skills.

4 Conclusion

My five-month internship at Sentient Scripts, facilitated by the AI-Driven Discovery Hub at SRM University, has been an invaluable learning experience. Throughout this period, I had the opportunity to work on challenging full-stack development projects, which allowed me to apply and expand my knowledge in Python, Next.js, and various AI tools and technologies.

Working in an environment that bridges academic research and industry application provided me with a unique perspective on how innovative technologies are developed and implemented. The support and guidance from both the AI-Driven Discovery Hub and Sentient Scripts were instrumental in my growth, helping me navigate complex problems and refine my technical skills.

This internship not only enhanced my technical abilities but also strengthened my problem-solving skills, adaptability, and understanding of real-world software development processes. The experience has been a significant milestone in my academic and professional journey, and I am confident that the skills and knowledge I have gained will serve me well in my future endeavors.

Completing these projects and contributing to the company's objectives has been immensely rewarding, and I am grateful for the opportunity to have worked with such a forward-thinking organization. This experience has solidified my interest in pursuing a career in software development and AI, and I am excited about the possibilities that lie ahead.

5 Appendix

– Nix Configuration File

```
{ pkgs ? import <nixpkgs> {} }:

let
  pythonPackages = pkgs.python3Packages;
  mmcv = pythonPackages.buildPythonPackage rec {
    pname = "mmcv";
    version = "1.7.2";

    src = pkgs.fetchPypi {
    inherit pname version;
    sha256 = "2116
a1609f9d3f59824074774366fd8b9410437bc95a24d535669b35d55352be"; #
Replace with the correct sha256 value for mmcv version 1.7.2
    };

    doCheck = false;

    propagatedBuildInputs = with pythonPackages; [
    pythonPackages.yapf
    pythonPackages.addict
    pythonPackages.opencv4
    ];

    meta = with pkgs.lib; {
    description = "OpenMMLab's collection of Computer Vision
tools.";
    homepage = "https://github.com/open-mmlab/mmcv";
    license = licenses.mit;
    maintainers = [ maintainers.your_maintainer_handle ];
    };
  };
in
pkgs.mkShell {
  buildInputs = [
    pythonPackages.matplotlib
    pythonPackages.torch
    pythonPackages.torchvision
    pythonPackages.setuptools
    pythonPackages.scipy
    pythonPackages.numpy
    pythonPackages.flask
    pythonPackages.flask-cors
    mmcv
  ];
}
```

– Dockerfile

```
FROM pytorch/pytorch

WORKDIR /code

RUN apt-get update && apt-get install ffmpeg libsm6 libxext6 -y

COPY api/ .
COPY public/image.png api/imgs

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

CMD python api/index.py
```